

POWER BI MINI PROJECT

Financial Sales Dashboard

Project Documentation Report

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1. Project Overview

This document presents the complete documentation for the Power BI Mini Project developed as part of the Agileology Data Analytics upskilling program. The project demonstrates end-to-end business intelligence capabilities using Microsoft Power BI Desktop, covering data ingestion, transformation, DAX modelling, and interactive dashboard development.

Attribute	Details
Project Title	Financial Sales Dashboard
Tool Used	Microsoft Power BI Desktop
Dataset	Financial_Sample.xlsx (Sheet1)
Total Records	700 Rows x 16 Columns
Dashboard Pages	4 Pages
DAX Measures	6 Measures + 1 Calculated Column
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2. Dataset Description

The dataset used for this project is the Microsoft Financial Sample dataset, a widely recognized business intelligence sample containing financial transaction data across multiple countries, product lines, and market segments.

2.1 Dataset Summary

Column Name	Data Type	Description
Segment	Text	Market segment (Government, Small Business, etc.)
Country	Text	Country of sale (USA, Canada, France, etc.)
Product	Text	Product name (Paseo, VTT, Amarilla, etc.)
Discount Band	Text	Discount category (High, Medium, Low, None)
Units Sold	Number	Number of units sold per transaction
Manufacturing Price	Currency	Cost to manufacture per unit
Sale Price	Currency	Selling price per unit

Column Name	Data Type	Description
Gross Sales	Currency	Revenue before discounts
Discounts	Currency	Total discount amount applied
Sales	Currency	Net sales after discounts
COGS	Currency	Cost of Goods Sold
Profit	Currency	Net profit (Sales minus COGS)
Date	Date	Transaction date
Month Number	Number	Month number (1-12)
Month Name	Text	Month name (January to December)
Year	Number	Year of transaction (2013-2014)

3. Data Preparation & Transformation

3.1 Data Import

The dataset was imported into Power BI Desktop using the Get Data > Excel Workbook option. The Sheet1 tab was selected in the Navigator window and loaded into Power Query Editor for transformation.

3.2 Power Query Transformations

- Verified all 700 rows and 16 columns were loaded correctly
- Renamed the Sales column (which had a leading space) to 'Sales' for clean referencing
- Verified data types: Date column set to Date, numeric columns set to Decimal Number, text columns set to Text
- Confirmed no null or duplicate rows existed in the dataset
- Applied Close & Apply to load the cleaned data into the data model

4. DAX Measures & Calculated Columns

4.1 Measures Table

A dedicated measures table named '_Measures' was created using the Enter Data option to organize all DAX measures separately from the source data table, following best practices in Power BI development.

4.2 DAX Measures Created

Measure Name	DAX Formula	Business Purpose
Total Sales	SUM('Sheet1'[Sales])	Total net revenue across all transactions
Total Profit	SUM('Sheet1'[Profit])	Total net profit after COGS deduction
Total Units Sold	SUM('Sheet1'[Units Sold])	Total units sold across all products
Total COGS	SUM('Sheet1'[COGS])	Total cost of goods sold
Total Discounts	SUM('Sheet1'[Discounts])	Total discounts given to customers
Profit Margin %	DIVIDE([Total Profit],[Total Sales],0)	Overall profitability percentage

4.3 Calculated Column

- Column Name: Profit Margin %
- Formula: `DIVIDE([Profit], [Sales]) * 100`
- Purpose: Calculates the profit margin percentage at the row level for each individual transaction

5. Dashboard Pages & Visualizations

Page 1 — Financial Overview

This page provides a high-level executive summary of the overall financial performance of the business.

Visual Type	Fields Used	Business Insight
KPI Card	Total Sales	Overall net revenue: 118.73M
KPI Card	Total Profit	Overall net profit: 16.89M
KPI Card	Total Units Sold	Total units sold: 1.13M
KPI Card	Total COGS	Total cost of goods: 101.83M
Clustered Bar Chart	Country vs Total Sales	USA and Canada lead with 25M each
Line Chart	Date vs Total Profit	Profit peaked at 2.03M in Nov 2014

Page 2 — Product & Segment Analysis

This page breaks down performance by individual product lines and customer segments.

Visual Type	Fields Used	Business Insight
Donut Chart	Segment vs Total Sales	Government dominates at 44.22% (52.5M)
Clustered Column Chart	Product vs Total Profit	Paseo leads profit at 4.8M
Matrix Table	Product, Total Profit, COGS, Sales, Units	Detailed product-level breakdown
Slicer	Year, Segment	Interactive filter for year and segment

Page 3 — Country & Discount Analysis

This page focuses on geographical performance and the impact of discount strategies on sales and profitability.

Visual Type	Fields Used	Business Insight
Stacked Bar Chart	Country, Segment vs Total Sales	USA leads in Government segment sales
Clustered Column Chart	Product vs Total Discounts	Paseo received highest discounts at 2.6M
Donut Chart	Segment vs Total Profit	Government contributes 65% of total profit
KPI Card	Profit Margin %	Overall margin: 19.53%
Slicer	Discount Band, Country	Filter by discount and geography

Page 4 — Yearly Trends & Performance

This page provides time intelligence analysis showing sales and profit trends over 2013 and 2014.

Visual Type	Fields Used	Business Insight
Line Chart	Month Name vs Total Sales by Year	October 2014 peaked at 12.4M in sales
Clustered Column Chart	Year vs Total Profit by Segment	2014 far outperformed 2013 across all segments
Area Chart	Month Name vs Sales and COGS	Sales consistently higher than COGS monthly
KPI Cards	Profit Margin %, Total COGS, Discounts	High-level performance summary
Slicer	Year (Dropdown)	Toggle between 2013 and 2014 data

6. Key Business Insights

- Total revenue of 118.73M was generated across 5 countries with a healthy profit of 16.89M (14.2% overall margin).
- Government segment is the highest contributor to both sales (44.22%) and profit (65.04%), making it the most valuable customer segment.
- Paseo is the top-performing product generating 4.8M in profit and 33M in total sales, significantly ahead of all other products.
- United States of America and Canada are the top two markets, each contributing approximately 25M in sales.
- Year 2014 showed significantly stronger performance than 2013 across all segments, especially in the Government category.
- October is consistently the highest sales month across both years, suggesting seasonal demand peaks.
- Despite high discount amounts given to Paseo (2.6M), it still generated the highest profit, indicating strong underlying product margins.
- Total COGS of 101.83M vs Total Sales of 118.73M confirms a lean operational model with controlled production costs.

7. Tools & Technologies Used

Tool / Technology	Purpose
Microsoft Power BI Desktop	Dashboard development, DAX modelling, and report creation
Power Query Editor	Data cleaning, transformation, and column renaming
DAX (Data Analysis Expressions)	Creating measures and calculated columns
Microsoft Excel (.xlsx)	Source dataset file format
Power BI Service	Publishing and sharing the dashboard online
GitHub	Version control and project portfolio hosting

8. Project Learnings & Skills Demonstrated

- Connecting Power BI Desktop to Excel data sources and navigating the Navigator window
- Performing data cleaning and transformation using Power Query Editor
- Creating a dedicated _Measures table following Power BI best practices
- Writing DAX measures including SUM, DIVIDE functions for business KPIs
- Building calculated columns using row-level DAX formulas
- Designing multi-page interactive dashboards with consistent theming
- Using KPI Cards, Bar Charts, Line Charts, Donut Charts, Area Charts, Matrix Tables, and Slicers

- Applying professional colour themes and formatting for executive-level reporting
- Publishing dashboards to Power BI Service for online access and sharing

9. Conclusion

This Power BI Mini Project successfully demonstrates the ability to build a complete, professional-grade business intelligence dashboard from raw data. The Financial Sales Dashboard provides meaningful insights across four analytical dimensions: overall financial health, product and segment performance, geographical and discount analysis, and time-based trends.

The project showcases practical skills in data ingestion, Power Query transformation, DAX formula development, multi-page report design, and Power BI Service publishing — all essential competencies for a Data Analyst or Power BI Developer role.

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